

Sai Nishchal Gamini

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Professional Experience

Google Cloud Infrastructure Engineer (GCP)

2021/12 – 2023/12

Infosys, Bengaluru

- Delivered client-focused technical support by diagnosing and resolving cloud infrastructure issues on Linux-based systems, with strong emphasis on IAM, logging, monitoring, and cloud migration scenarios.
- Investigated and mitigated incidents by analyzing logs and system metrics (e.g., CPU, memory, disk, network) across Compute Engine Linux VMs and Kubernetes clusters, ensuring minimal downtime for production workloads.
- Collaborated with clients and cross-functional SMEs to design and execute migrations from on-premise Linux environments to cloud infrastructure, supporting use cases such as web application hosting, batch processing, and containerized microservices.
- Enhanced system reliability by proactively monitoring Linux-based infrastructure, optimizing IAM access controls, and improving observability practices across both legacy and migrated cloud environments.

Cloud Infra trainee

2021/09 – 2021/12

- Completed hands-on cloud training via Qwiklabs, gaining experience in IAM, Compute Engine, Kubernetes, Linux, monitoring, and Terraform for infrastructure as code.
- Applied learning in simulated environments to implement cloud migrations from on-premise systems, successfully deploying and managing production-like infrastructure.

Education

Master's Digital Engineering (Computer Science)

2024/04 – present

Otto-von-Guericke-Universität

Magdeburg, India

Bachelor's Mechanical Engineering

2017/07 – 2021/07

SRKR Engineering College

Bhimavaram, India

Skills

AI&NLP: RAG (Retrieval-augmented generation), Embeddings & vector representations, Question answering, Summarization, Multimodal NLP, Text classification, Intent classification

Databases: MySQL, MongoDB, Milvus, Cloud Bigtable, Cloud Firestore,

Programming Languages: Python, Java11, C++, Bash, SQL

ML/Data Science Libraries: PyTorch, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, SciPy, Joblib, ImageIO, NetworkX, Hugging Face, Transformers, LangChain

XAI Techniques: Grad-CAM, Latent Space Traversals, t-SNE, DiCE, Reconstruction Error Heatmaps

Backend & Microservices: Spring Boot, Spring Data, RESTful API Development, Node.js, Flask

Frontend & WebDev: React, JavaScript, HTML5, CSS3, Bootstrap 4

Cloud technology and tools : Google cloud platform(GCP), BigQuery, Cloud Dataflow, Dataproc, Cloud monitoring

DevOps&tools: Docker, Git, Kubernetes, Microsoft Excel, Linux

Projects

Multimodal Conversational AI

- Developed an NLP-focused multimodal RAG conversational AI system with intent classification, semantic retrieval (dense and graph-based), and LLM-based generation. Created the Chatbot frontend with React and JavaScript, Integrated Milvus for embedding storage and STT/TTS for spoken and text interactions, achieving 90% retrieval accuracy and improving to 99% via graph-based diffusion re-ranking with low-latency performance.

Data Analysis & Visualization of Persistent Memory Systems (Optane PMem)

- Analyzed a structured dataset of 261 research studies to identify performance trends and evaluation gaps, deriving actionable recommendations through benchmark and metric comparison. Developed meaningful visualizations and frameworks to present insights clearly, and collaborated within a team to coordinate analysis tasks and communicate results effectively.

Understanding deep learning model with XAI

- Developed and deployed a disentangled Variational Autoencoder (VAE) with a CNN-based architecture, integrating multiple explainable AI (XAI) techniques for model interpretability. Trained and evaluated the model in a Dockerized environment, achieving 100% test accuracy in category prediction and demonstrating high reliability and robustness.

Certificates

Python data structures | Image processing in python | Deep Learning for Images with PyTorch |
Explainable Artificial Intelligence (XAI) Concepts | Data structures and algorithms using C++ |
Google cloud associate engineer

Papers

Conference on Computational Linguistics (CLiC-it 2025)

Diffusion-Aided RAG: Elevating Dense-Retrieval Chatbots via Graph-Based Diffusion Reranking

Languages

- English(C1)
- German(A2)

Organizations

Academic club FIN, OVGU

Assitant Student Lead

2024/11 – 2025/10
Magdeburg, Germany

- Organized and hosted academic events, workshops, and hackathons (*Thesis Talk, Unity Workshop*) to engage students with emerging technologies and industry trends.
- Collaborated with industry partners and institutions to deliver interactive learning experiences and enhance student exposure to real-world applications.